



Ahmedabad
University

ECE 310 – Wireless Communications

Milestone 1 (M1): Proposal & Scoping

Project Category: Minimum Viable Product (MVP)

ElderGuard: A Low-Cost Wi-Fi CSI Sensing System for Fall Detection in Elderly Care

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Section: Group 8

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1. Problem Statement

Falls are a serious and common problem for people living alone, especially the elderly. The World Health Organization lists falls as the second leading cause of unintentional injury death worldwide, with an estimated 684,000 deaths a year and over 37 million falls annually severe enough to need medical attention; adults over 60 make up the largest share of fatal falls. How quickly a fall is detected directly affects the outcome for someone living alone.

Existing fall-detection approaches are wearables, cameras, or device-free ambient sensing. Wearables get forgotten or feel uncomfortable, especially in the bathroom, one of the most common fall locations. Cameras raise privacy concerns in bedrooms and bathrooms. This is why we are building on Wi-Fi Channel State Information (CSI) sensing: it needs no device on the body and does not capture an image, only how a person's movement disturbs an existing Wi-Fi link.

The specific wireless communication problem: when a person moves between a Wi-Fi transmitter and receiver, their body reflects, scatters, and diffracts the signal, producing measurable amplitude and phase variation across OFDM subcarriers. A fall has a distinctive Doppler signature: a large, sudden displacement followed by stillness. The challenge is separating that signature from hardware phase noise, static multipath from the room itself, and other everyday activities, while also knowing when the underlying wireless link is too poor (low SNR, high packet loss) to trust the detection at all.

2. Target User

- Primary user: an elderly person living alone or with a part-time caregiver, who cannot or will not consistently wear a fall-detection pendant or smartwatch.
- Secondary user: a family member or caregiver who receives the fall alert on their phone and needs it to be fast and trustworthy enough to act on.
- Tertiary/future user: small eldercare facilities that want room-level fall monitoring without installing cameras.

3. Product Concept

ElderGuard is a pair of low-cost ESP32-S3 boards placed in a room like a normal piece of home networking equipment: one acts as a Wi-Fi transmitter, the other as a receiver. Instead of connecting to a camera or asking the person to wear anything, the boards continuously capture Channel State Information from the Wi-Fi link between them. This CSI is cleaned, filtered, and converted into a Doppler spectrogram, which a compact CNN classifies as a fall, normal activity, or no activity. Alongside the classification, the system reports a live confidence score based on the current signal-to-noise ratio and packet delivery rate, so the caregiver knows not just that a fall was flagged, but how much to trust that specific flag.

At M1, this pipeline works on raw filtered amplitude. From M2 onward, we plan to move to a body-coordinate velocity profile (BVP)-style feature, following the approach used in Widar 3.0, which represents motion as speed

and direction rather than raw signal shape. This representation is far less tied to a specific room's walls and furniture, which is what directly targets the cross-environment generalization gap identified in Section 7. We also plan to classify the full 2D spectrogram with a small CNN plus a lightweight subcarrier-attention layer, instead of averaging all 52 subcarriers into a single line as the current M1 code does.

4. Value Proposition

- No wearable to remember, charge, or take off in the shower — unlike pendant-style medical alert devices.
- No camera and no stored images or video — addresses the privacy objection that stops many families from installing camera-based monitoring in bedrooms and bathrooms.
- Built on commodity, low-cost Wi-Fi hardware (ESP32-S3 boards, roughly the cost of a smart plug) rather than specialized radar or depth-camera sensing hardware.
- Reports its own confidence rather than presenting every detection as equally certain, which is something we did not find in the fall-detection literature we reviewed.

5. Functional Requirements

- Classify the current activity as fall, normal activity, or no activity, in real time, from live CSI.
- Raise an alert within 500 ms of a real fall event on the physical testbed.
- Compute and display a live confidence score derived from measured SNR and packet delivery rate.
- Log detection events (timestamp, classification, confidence score) for later review.
- Operate using two ESP32-S3 boards capturing CSI (I/Q across 52 subcarriers, RSSI, timestamp) at roughly 100 Hz over an 802.11n, 20 MHz link.

6. Non-Functional Requirements

- Privacy: no images, audio, or biometric identity data are captured or stored at any point.
- Cost: total hardware cost kept low by using off-the-shelf ESP32-S3 boards rather than specialized radar/depth sensors.
- Latency: end-to-end detection-to-alert time under 500 ms, matching the functional requirement above.
- Robustness: usable Wi-Fi signal is unaffected in a single small room without the person needing to stand in a specific spot.
- Portability: the pipeline should degrade gracefully — and be explicitly measured — when moved to a room it was not trained in, rather than failing silently.
- Confidence: from M2 onward, the reported confidence score should combine the model's own calibrated uncertainty with link-quality metrics, not link quality alone.

7. Review of Existing Work and SOTA Anchor

We grouped the related work into CSI-specific fall/activity classifiers and cross-domain generalization methods. The table below identifies the most relevant work and the specific gap each leaves for a home elder-care MVP.

Work	Method	Reported Result	Gap Relative to ElderGuard
Chu et al., IEEE Access 2023	CNN on CSI reshaped as an image, 4 indoor rooms	Over 96% cross-environment accuracy	No reported link-quality (SNR/PDR) breakdown; only 4 self-collected rooms
Lightweight CNN benchmark (MobileNetV2/V3, EfficientNet-B0, ShuffleNetV2)	Compares 4 edge-friendly CNNs	EfficientNet-B0 best: 98.61% accuracy, 8.53 ms inference	Optimized for edge efficiency, not cross-environment/cross-person generalization
Cai et al., KNN-MMD (cross-domain Wi-Fi sensing)	Few-shot local distribution alignment across rooms	Outperforms standard domain-adaptation baselines	Needs labelled samples from the new room; not plug-and-play

SOTA anchor for M1: for raw classification accuracy, we treat Chu et al. (IEEE Access, 2023) — a CNN-based CSI fall classifier reporting over 96% cross-environment accuracy — as the closest existing result to what we are building. For the generalization problem specifically, we treat the KNN-MMD few-shot cross-domain line of work as the current state of the art, since it directly targets the accuracy drop that plain CNN classifiers show when moved to a new room. Neither line of work reports a link-quality-aware confidence score, which is the gap our MVP is aimed at. This anchor is a starting point and will be repositioned at M2 once we have our own results on a public dataset to compare against.

Concretely, closing this gap means our M2 model should not just be a plain amplitude-plus-CNN classifier. We plan to build it around the BVP-style feature and subcarrier attention described in Section 3, and to report its confidence as a calibrated model output (e.g. via Monte Carlo Dropout or a predicted probability) combined with link quality, rather than link quality alone. At M2 we will benchmark this against a plain amplitude+CNN baseline on the same data split, so the comparison itself becomes evidence that the added complexity is worth it.

8. Wireless Communication Concepts Involved

As required for any project in this course, the table below makes explicit which ECE 310 concepts the project is built on, and where each one actually appears in the system rather than just being mentioned.

ECE 310 Concept	Where It Shows Up in ElderGuard
Multipath propagation & fading	CSI amplitude/phase variation across subcarriers is caused by multipath reflecting off the person and the room
Doppler shift	A fall produces a short, high-Doppler burst in the CSI time series; this is what the spectrogram is built to capture
OFDM / subcarriers	CSI is measured per-subcarrier (52 subcarriers, 802.11n, 20 MHz); the pipeline processes each subcarrier before combining them
SNR / SINR and link quality	Feeds the live confidence score: a fall decision made on a low-SNR, high-packet-loss link is flagged as less trustworthy
Signal representation (amplitude & phase)	The phase-sanitization step removes CFO/timing-offset drift so the remaining phase reflects real motion, not hardware artifacts
Noise and interference	The 0.5–10 Hz Butterworth bandpass filter is designed specifically to reject noise and interference outside the human-movement band

9. Evaluation Metrics (defined before experimentation)

Metric	What It Tells Us	Target
F1-score (fall class)	Balances missed falls against false alarms; more meaningful than raw accuracy since falls are rare events	As high as possible; prioritized over plain accuracy
False Negative Rate	Missed falls are the most costly error in this application	Kept as low as possible, even if it costs some false positives
End-to-end latency	Time from the physical fall to the alert being raised	Under 500 ms on the hardware testbed
F1-score vs. SNR / packet delivery ratio	Shows how much link quality affects detection reliability; ties results back to wireless theory	Report the trend, not a single number
Cross-environment F1 drop	Difference in F1-score between a room the model has seen and one it hasn't (leave-one-environment-out)	As small as possible; this is our generalization claim

10. Technical Progress at M1

Since the September 5 proposal, we scaffolded the GitHub repository (/data_pipeline, /ml_models, /evaluation, /hardware, plus root README) and implemented a first working version of the signal-processing stage: CSI phase sanitization (unwrap plus linear-fit removal of timing/frequency offset), a 4th-order Butterworth bandpass filter over 0.5–10 Hz, and STFT-based Doppler spectrogram generation. This has been validated end-to-end on a synthetic CSI-like signal (a fall-like burst against a quiet background), and correctly produces a clear, isolated burst of movement energy exactly where the synthetic fall occurs. We also shortlisted Widar 3.0 and UT-HAR as candidate public datasets and drafted the leave-one-person/environment-out split logic the eventual model will be evaluated with, and identified the ESP32-S3 CSI acquisition path (the open-source ESP32 CSI Tool / Espressif CSI stack) for the Stage 2 hardware testbed.

11. Timeline

Milestone	Date	Planned MVP Deliverable (per course rubric)
M1 (this submission)	13 Sep 2026	Problem, target user, product concept, value proposition, functional/non-functional requirements, SOTA anchor
M2	11 Oct 2026	System architecture, technology stack, implementation plan, revised SOTA positioning, one working core function (the preprocessing pipeline running on real public-dataset CSI)
M3	01 Nov 2026	Prototype description, requirements coverage, test results, limitations, revised SOTA positioning (includes ESP32-S3 hardware bring-up)
M4	22 Nov 2026	Final MVP, user documentation, setup instructions, evaluation against M1 requirements, limitations, final SOTA positioning

12. Team and Individual Contributions

- **Falak Shah (AU2440014)** — Defined the problem statement, target user, and product concept; wrote the CSI signal-processing pipeline (phase sanitization, bandpass filtering, Doppler spectrogram) and validated it on synthetic data.
- **Hiranshee Doshi (AU2440228)** — Led the related-work review and SOTA anchor: identified and compared Chu et al., the lightweight-CNN benchmark, and the KNN-MMD cross-domain work.
- **Nikunj Kalariya (AU2440275)** — Defined functional and non-functional requirements and the value proposition; built the wireless-concepts mapping table connecting ECE 310 theory to the project.
- **Aaryan Parmar (AU2440229)** — Defined the evaluation metrics (F1-score, false negative rate, latency, SNR/PDR correlation, cross-environment F1 drop) and drafted the M1–M4 timeline against the official milestone dates.
- **Priyanshi Verma (AU2440171)** — Researched and shortlisted the public datasets (Widar 3.0, UT-HAR) and the ESP32-S3 CSI acquisition path; set up the GitHub repository structure.

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